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S32K312MINI-EVB

Mass-Market Board

CAUTION :

This schematic is provided for reference purposes only. As such, NXP does not make any warranty, implied or otherwise, as to the suitability of circuit design or component selection (type or value) used in these schematics for hardware design using the NXP S32K family of Microprocessors. Customers using any part of these schematics as a basis for hardware design, do so at their own risk and NXP Semiconductors does not assume any liability for such a hardware design.

3 Different test points used in design:

TPVx - Through Hole Pad small

TPHx - Through Hile Pad Large (for standard 0.1" header). Also used on IO Matrix (IOMx)

TPX - Surface Mount Wire Loop

TPV?

TPH5

TP?

Notes :

- All components and board processes are to be ROHS compliant
- All connectors and headers are denoted Jx/Px and are 2.54mm pitch unless otherwise stated
- All jumpers are denoted Jx. Jumpers are 2mm pitch
- Jumper default positions are shown in the schematics. For 3 way jumpers, default is always posn 1-2.
- 2 Pin jumpers generally have the "source" on pin 1.
- All switches are denoted SWx
- All test points (SMT wire loop style) are denoted TPx
- Test point Vias (just through hole pads) are denoted TPVx

Signals (ports) have not been routed via busses as this makes it harder to determine where each signal goes.

User notes are given throughtout the schematics.

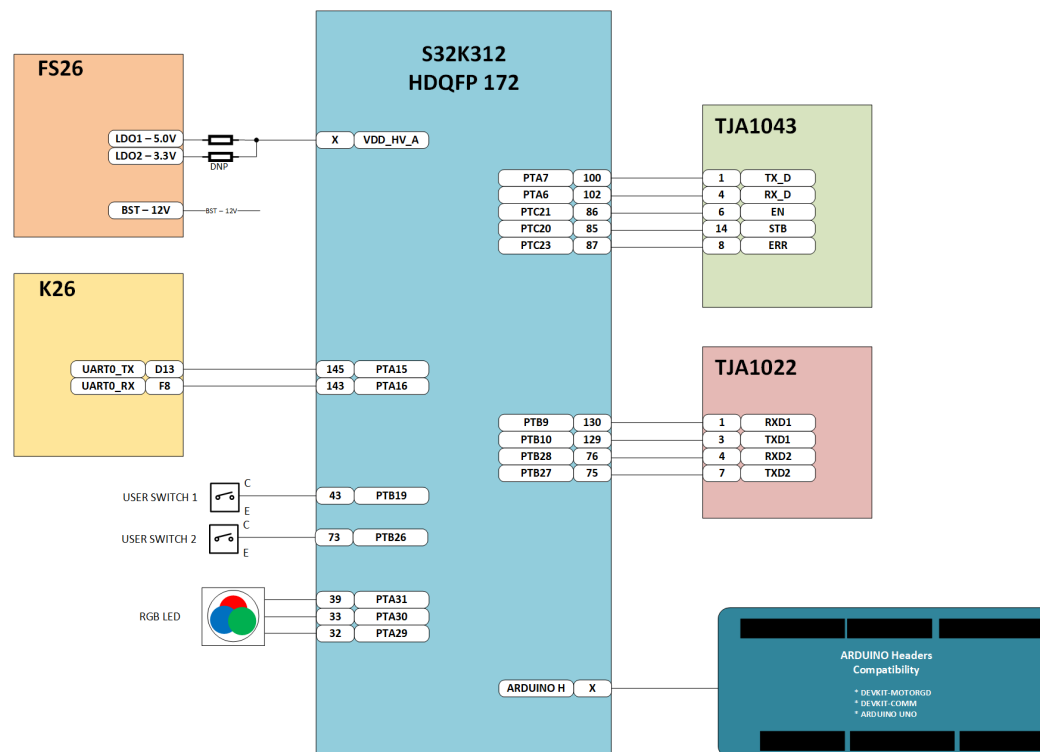
Specific PCB LAYOUT notes are detailed in ITALICS

Revision History			
Rev	Description	Date	Approved/Designer
X0	A055	26/06/2024	Luis Rico
X1	Motor Control added, S32K312 symbol updated, CAN, LIN connection updated, compatibility DEVKIT-COMM & MotorKit. FS26 DEBUG mode update, Arduino H update, notes for CAD added in Arduino H Replacements added, LED defined, FS26 feedback VDDIO	24/07/2024	Luis Rico
A	A085	16/08/2024	Luis Pablo Melano
B	RESET signal disconnected from FS26, SW3 added, C81 --> 1uF, R117--> 4.7M Table silk added, SW3 added SPI00 connection at Arduino H Pilot Version	16/12/2024	Luis Rico
C	Changes just in layout, Rev added to be paired ith LAY	22/01/2025	Luis Pablo Melano

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Designer: Luis Rico	Drawing Title: S32K312MINI-EVB		
Drawn by: Luis Rico	Page Title: TITLE PAGE		
Approved: Jesus Sanchez	Size B	Document Number SCH-94288 PDF: SPF-942888	Rev C
Date: Wednesday, January 22, 2025		Sheet 1 of 12	

REF DES	LINKER(DEFAULT)	PAGE NAME
U19, J20	F:2	D3, FS26

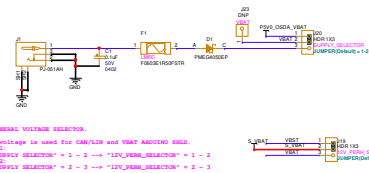
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U29	DNP	D4, MCU PWR
E116, C117, J18, R38, R92	DNP	D9, CANLIN PHY
R111, R118, R176	DNP	D, OPENSDA K26
R120, R122, R124, R126, R128, R130, R133, R135, R136, R137, R151, R157, R161, R164, R166, R168	DNP	F1, ARDUINO HDRS



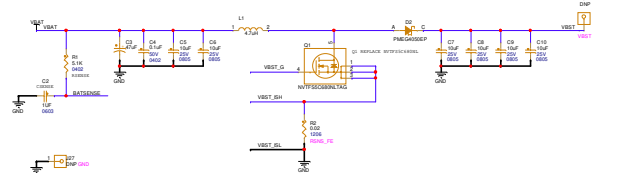
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Doc ID: 44133 Doc Rev: 1		Product Name: S32K312MINI-EVB	Classification: E2P
NOTES & BLOCK DIAGRAM			
Approved: Jeroen Grooten		Date: 2023-08-01	Approved: SCHWABER PROF. SPT-2023-08-01

FS26 - Power Supply

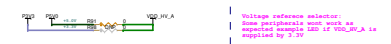
Supply selector



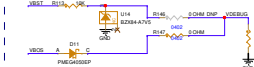
Boost +12V



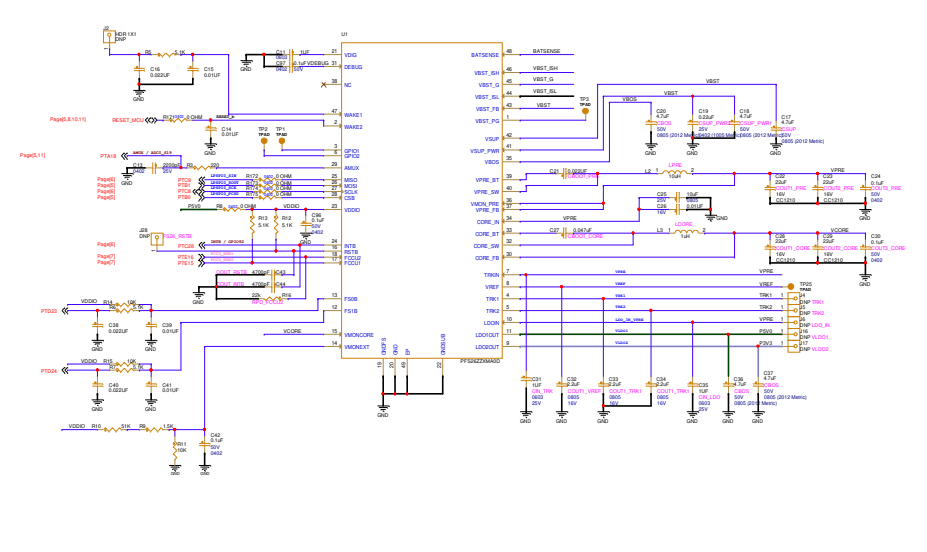
Voltage references



DEBUG



SMPS-based SBC



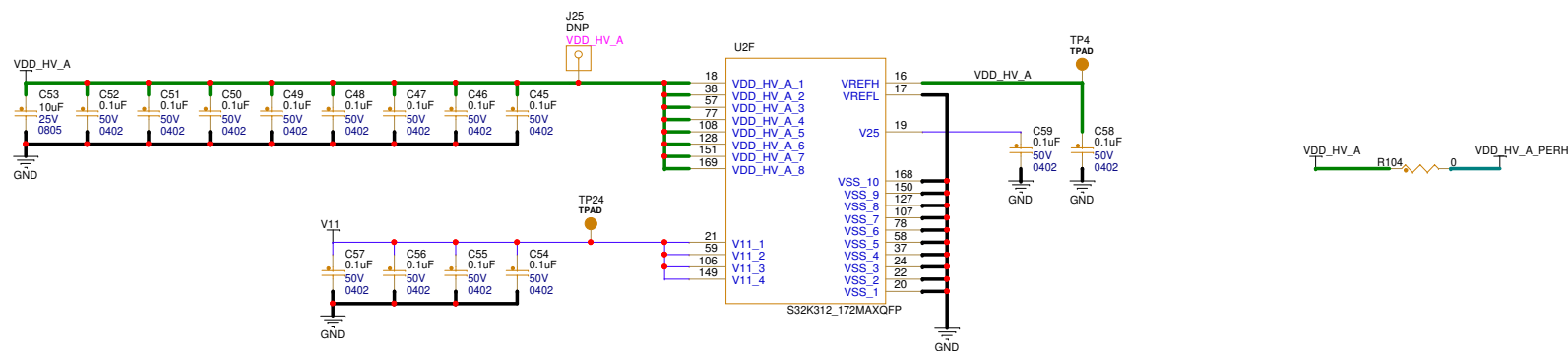
Use the following test as Silk

Conn	USB-C	JACK
J19	1 - 2	2 - 3
J20	1 - 2	2 - 3

Domain	Conn
VDD_WF_A	J20
VDD01	J16
VDD02	J17
GND	J27

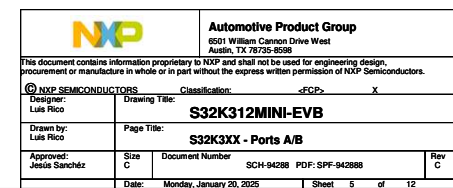
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Doc No:	FS26	Doc Rev:	1.0
Doc Title:	FS26 SBC Power Supply	Doc Type:	Block Diagram
Doc Author:	SCHWAB, POF-SPT-000000	Doc Date:	2023-09-01

Power Supply Pins



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Drawn by: Luis Rico	Page Title: S32K3XX - MCU POWER Supply Pins		
Approved: Jesús Sánchez	Size B	Document Number SCH-94288 PDF: SPF-942888	Rev C
Date: Monday, January 20, 2025		Sheet 4 of 12	

Ports A/B



S32K312 Microcontroller - 172HDQFP

Ports C/D

		U2C
Page[11]	PTC0	PTC0/GPIO6/CAN3_TX/EMIOS_0_CH0_X/EMIOS_0_CH14_H/EIRQ0
Page[11]	PTC1	PTC1/GPIO6/EMIOS_0_CH1_G/FXIO_05/EMIOS_0_CH15_H/CAN3_RX/EIRQ1
		PTC2/GPIO6/EMIOS_0_CH2_G/LPSPI0_PCS0/LPSPI0_PCS3/CAN0_RX/LPUART0_RX/EIRQ2/CMPO_IN2
		PTC3/GPIO6/EMIOS_0_CH3_G/CAN0_TX/LPUART0_TX/SC0_PUR/EIRQ3/CMPO_IN4
		PTC4/GPIO6/EMIOS_0_CH4_X/FXIO_D5/UTAG_TCK/SWD_CLK/EIRQ4/CMPI_IN3
		PTC5/GPIO6/EMIOS_0_CH16_X/FXIO_D4/UTAG_TDIL/P2C1_HREQ/EIRQ5/ADC1_S14
Page[10]	JTAG_TCLK/SWD_CLK	PTC6/GPIO7/LP2C1_SDA/FXIO_D11/LPSPI1_PCS1/LCU0_OUT/EMIOS_1_CH6_H/LPSPI0_PCS1/ADC0_MA2/CAN2_RX/LPUART1_RX/EIRQ6/WKPUS/ADC1_S18
Page[10]	JTAG_TDI	PTC7/GPIO7/FXIO_D10/LPUART1_TX/LP2C1_SCL/LCU0_OUT/EMIOS_1_CH7_H/LPSPI0_PCS0/CAN2_TX/LP2C0_HREQ/EIRQ7/WKPUS/ADC1_S16
Page[11]	PTC7	PTC8/GPIO7/LP2C0_SCL/CAN1_TX/EMIOS_1_CH8_H/LCU1_OUT7/LPSPI0_SCK/FXIO_D12/LPUART0_CTS/LPUART1_RX/EIRQ16/ADC0_S12
Page[11]	PTC8	PTC9/GPIO7/LP2C0_SDA/LPUART1_TX/LPUART0_RTS/EMIOS_1_CH9_X/LCU1_OUT8/LPSPI0_SIN/FXIO_D13/CAN1_RX/EIRQ17/ADC0_S13/WKPUS10
Page[10]	PTC9	PTC10/GPIO7/EMIOS_0_CH8_G/CAN5_TX/LPSPI2_PCS1/LCU1_OUT11/EMIOS_1_CH0_X/EIRQ18/TRGMUX_IN11/ADC1_X3
Page[11]	PTC10	PTC11/GPIO7/EMIOS_1_CH1_H/FXIO_D15/FXIO_D16/LCU1_OUT10/CAN5_RX/EIRQ19/TRGMUX_IN10/WKPUS18/ADC0_S17
Page[11]	PTC11	PTC12/GPIO7/ADC1_MA2/EMIOS_1_CH2_H/EMIOS_0_CH2_X/LPSPI2_PCS1/FXIO_D19/LCU1_OUT9/EIRQ20
Page[11]	PTC12	PTC13/GPIO7/EMIOS_1_CH3_H/EMIOS_0_CH23_X/ADC1_MA1/FXIO_D16/LCU1_OUT6/EIRQ21
Page[11]	PTC13	PTC14/GPIO7/EMIOS_0_CH10_H/LPSPI0_PCS0/ADC0_MA1/EMIOS_1_CH4_H/LCU1_OUT11/FXIO_D16/CAN2_RX/EIRQ22/TRGMUX_IN9
Page[11]	PTC15	PTC15/GPIO7/CAN2_TX/EMIOS_0_CH11_H/LPSPI2_SCK/ADC0_MA2/LPUART2_TX/LCU1_OUT0/LP2C1_SCL/EIRQ23/TRGMUX_IN8
Page[11]	PTC16	PTC16/GPIO8/LPSPI0_SIN/EMIOS_1_CH5_H/LP2C1_SDA/S3C0_SCL/FXIO_D15/LP2C1_SCL/CAN2_RX/LPUART2_RX
Page[11]	PTC18	PTC17/GPIO8/LPSPI0_SCK/SC0_PUR/CAN2_TX/LP2C1_SCL/S/FXIO_D14/FXIO_D2
Page[11]	PTC19	PTC18/GPIO8/FXIO_D6/FXIO_D15/LCU1_OUT7/LPUART6_RX/WKPUS6
Page[11]	PTC20	PTC19/GPIO8/LPUART7_TX/FXIO_D13/LP2C1_PCS1/LCU1_OUT8
Page[11]	PTC21	PTC20/GPIO8/FXIO_D14/ADC1_MA2/LCU1_OUT5/LPUART7_RX/EIRQ16/WKPUS43
Page[11]	PTC22	PTC21/GPIO8/FXIO_D15/ADC1_MA1/LCU1_OUT4/EIRQ17
Page[11]	PTC23	PTC23/GPIO87/FXIO_D16/LCU1_OUT0/EIRQ18/WKPUS44
Page[11]	PTC24	PTC24/GPIO8/EMIOS_1_CH6_X/FXIO_D17/LCU1_OUT11/TRGMUX_OUT15/EIRQ19/WKPUS46
Page[11]	PTC25	PTC25/GPIO8/EMIOS_1_CH1_H/FXIO_D18/LCU1_OUT2/EIRQ20/ADC0_S20/WKPUS45
Page[11]	PTC26	PTC26/GPIO9/EMIOS_1_CH3_H/FXIO_D19/LCU1_OUT9/CAN5_RX/EIRQ21/ADC0_S21/WKPUS48
Page[11]	PTC27	PTC27/GPIO9/CAN5_TX/EMIOS_1_CH4_H/FXIO_D20/ADC1_MA3/LCU1_OUT3/EIRQ22
Page[11]	PTC28	PTC28/GPIO9/CAN3_TX/EMIOS_1_CH7_H/FXIO_D21/FXIO_D22/LP2C1_SCL/LCU1_OUT8/SC0_SCL
Page[11]	PTC29	PTC29/GPIO8/EMIOS_1_CH10_H/FXIO_D22/LP2C1_SDA/ADC0_SDA/FXIO_D3/CAN3_RX/EIRQ23/WKPUS47
Page[11]	PTC30	PTC30/GPIO8/CAN4_TX/EMIOS_1_CH12_H/FXIO_D19/SC0_PUR/FXIO_D23
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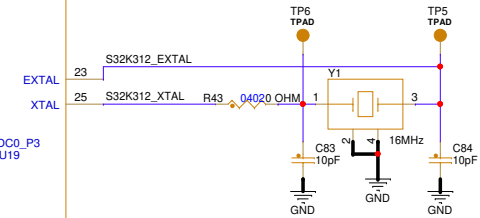
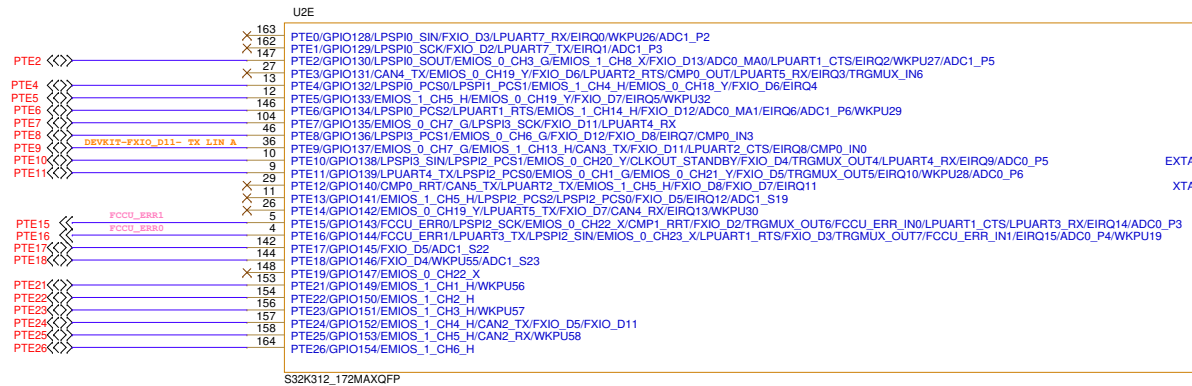
S32K312_172MAXQFP

		U2D
Page[11]	PTD0	PTD0/GPIO9/EMIOS_0_CH2_G/LPSPI0_SOUT/EMIOS_0_CH16_X/FXIO_D0/TRGMUX_OUT11/LPUART5_RX/EIRQ26/WKPUS/ADC0_P1
Page[11]	PTD1	PTD1/GPIO27/LPUART5_TX/EMIOS_0_CH3_G/LPSPI0_SCK/EMIOS_0_CH17_Y/FXIO_D1/TRGMUX_OUT5/EIRQ28/ADC0_P0
Page[11]	PTD2	PTD2/GPIO28/LCU0_OUT1/EMIOS_1_CH21_Y/LPSPI1_SOUT/FXIO_D4/FXIO_D6/LPUART3_TX/EIRQ10/TRGMUX_IN5/WKPUS/ADC0_S16
Page[11]	PTD3	PTD3/GPIO28/EMIOS_1_CH22_X/LPSPI1_PCS0/FXIO_D5/FXIO_D7/LCU0_OUT0/LPUART3_RX/EIRQ11/TRGMUX_IN4/NMI_B/WKPUS1/ADC0_S10
Page[11]	PTD4	PTD4/GPIO100/EMIOS_1_CH23_X/LPSPI1_PCS1/LCU0_OUT8/EIRQ12/ADC0_S18/WKPUS22
Page[11]	PTD6	PTD5/GPIO101/EMIOS_0_CH19_Y/EMIOS_0_CH2_G/SC0_SDA/FXIO_D15/LPSPI0_PCS1/EIRQ13/TRGMUX_IN7
Page[11]	PTD7	PTD6/GPIO102/FXIO_D3/EMIOS_1_CH12_H/FXIO_D15/LPSPI0_PCS1/LPUART2_RX/EIRQ14/CMPO_IN7
Page[11]	PTD8	PTD7/GPIO103/LPUART2_TX/LPSPI0_PCS3/LPSPI0_PCS3/EIRQ15/CMPO_IN6
Page[11]	PTD9	PTD8/GPIO104/LPSPI0_SOUT/LP2C1_SDA/SC0_SDA/FXIO_D1/EMIOS_0_CH12_H/FXIO_D11/LPUART6_RX/EIRQ24
Page[11]	PTD10	PTD9/GPIO105/LP2C1_SCL/FXIO_D0/LPUART6_TX/SC0_SCL/EMIOS_0_CH13_H/FXIO_D10/EIRQ25
Page[11]	PTD11	PTD10/GPIO106/EMIOS_0_CH16_X/EMIOS_1_CH10_H/LPSPI0_SIN/CLKOUT_RUN/EIRQ26
Page[11]	PTD12	PTD11/GPIO107/EMIOS_0_CH17_Y/LPSPI0_SCK/LPUART2_CTS/EIRQ27
Page[11]	PTD13	PTD12/GPIO108/EMIOS_0_CH18_Y/LPUART2_RTS/LPSPI0_SOUT/LP2C1_HREQ/EIRQ28
Page[11]	PTD14	PTD13/GPIO109/EMIOS_0_CH20_Y/FXIO_D7/LP2C0_SDA/LPUART1_RX/EIRQ29/WKPUS4
Page[11]	PTD15	PTD14/GPIO110/EMIOS_0_CH21_Y/LPUART1_TX/LP2C0_SCL/CMPO_RTT/CLKOUT_RUN/EIRQ30
Page[11]	PTD16	PTD15/GPIO111/FXIO_D6/EMIOS_0_CH3_X/EMIOS_1_CH14_H/LPSPI0_SCK/FXIO_D1/CAN3_RX/LPUART2_CTS/EIRQ31/CMPO_IN1
Page[11]	PTD17	PTD16/GPIO112/EMIOS_0_CH1_G/LPSPI0_SIN/EMIOS_1_CH15_H/LPUART2_RTS/CMPO_IN5
Page[11]	PTD18	PTD17/GPIO113/EMIOS_0_CH16_H/LPSPI0_PCS0/FXIO_D9/CAN5_RX/LPUART2_RX/EIRQ24
Page[11]	PTD19	PTD18/GPIO114/CAN2_TX/EMIOS_1_CH15_H/FXIO_D2
Page[11]	PTD20	PTD20/GPIO115/EMIOS_1_CH17_Y/FXIO_D26/LPSPI1_PCS2/LPSPI0_SIN/EIRQ25/ADC0_S22/WKPUS4
Page[11]	PTD21	PTD21/GPIO116/EMIOS_1_CH18_Y/FXIO_D28/LCU0_OUT4/EIRQ26/ADC0_S23
Page[11]	PTD22	PTD22/GPIO118/EMIOS_1_CH19_Y/FXIO_D27/LCU0_OUT5/EIRQ27
Page[11]	PTD23	PTD23/GPIO119/EMIOS_1_CH20_Y/FXIO_D28/HSE_TAMPER_LOOP_OUT0_LCU0_OUT10/EIRQ28/WKPUS0
Page[11]	PTD24	PTD24/GPIO120/EMIOS_1_CH21_Y/FXIO_D29/LCU0_OUT11/HSE_TAMPER_EXTN0/EIRQ29/ADC1_S20
Page[11]	PTD25	PTD25/GPIO122/EMIOS_1_CH23_X/FXIO_D7/FXIO_D30/LCU0_OUT0
Page[11]	PTD26	PTD26/GPIO123/FXIO_D31/LCU0_OUT11/EIRQ30/ADC1_S21/WKPUS1
Page[11]	PTD27	PTD28/GPIO124/LCU0_OUT2/EIRQ31/ADC1_S22
Page[11]	PTD28	PTD29/GPIO125/LCU0_OUT3/ADC1_S23/WKPUS2
Page[11]	PTD29	PTD30/GPIO126/LCU0_OUT6
Page[11]	PTD30	PTD31/GPIO127/FXIO_D6/LCU0_OUT9/WKPUS3

S32K312_172MAXQFP

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Drawn by: Luis Rico	Page Title: S32K3XX - Ports C/D		
Approved: Jesse Sanchez	Size: C	Document Number: SCH-94288 PDF: SPF-942888	Rev: C
Date: Monday, January 20, 2025		Sheet: 6	of 12

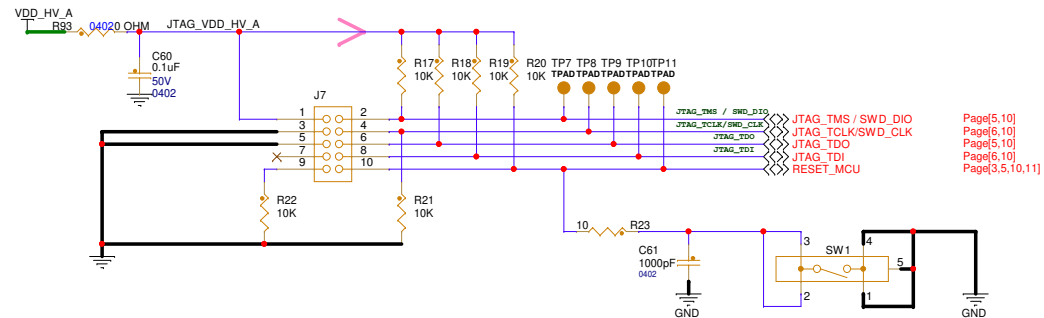
S32K312 Microcontroller - 172HDQFP Port E



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Approved: Jesús Sánchez	Size B	Document Number SCH-94288 PDF: SPF-942888	Rev C
Date: Monday, January 20, 2025		Sheet 7 of 12	

JTAG Connector

10-pin Cortex Debug Connector

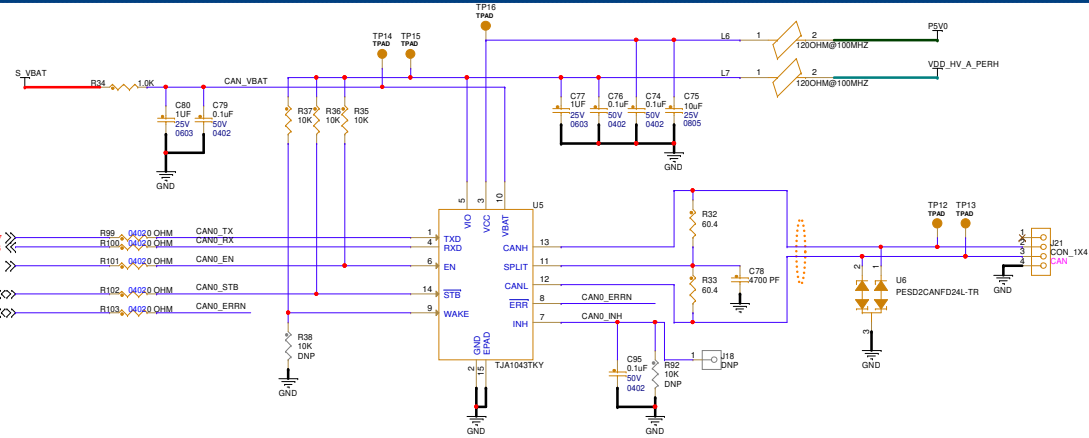


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Date: Monday, January 20, 2025		Sheet 8 of 12	

CAN / LIN PHY

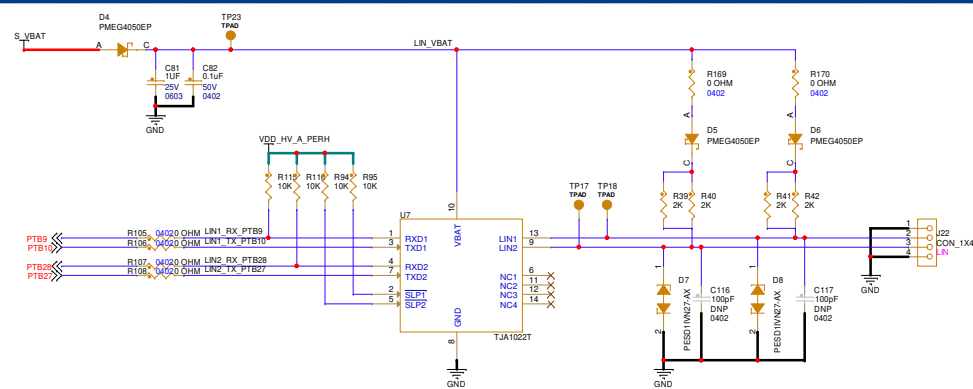
TJA1043 - CAN PHY

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TJA1022 - DUAL LIN PHY

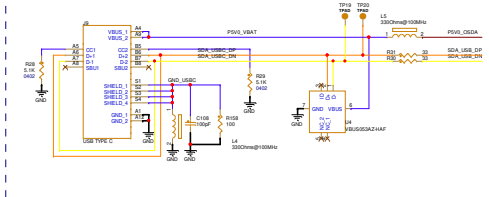
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Drawn by: Luis Rico	Page Title: CAN/LIN PHY		
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Date: Monday, January 20, 2025		Sheet 9 of 12	

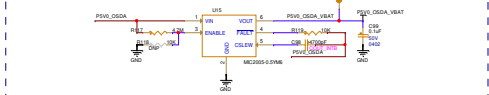
MK26 OpenSDA

USB-C Connector

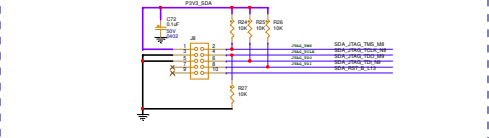


Current limiting USB

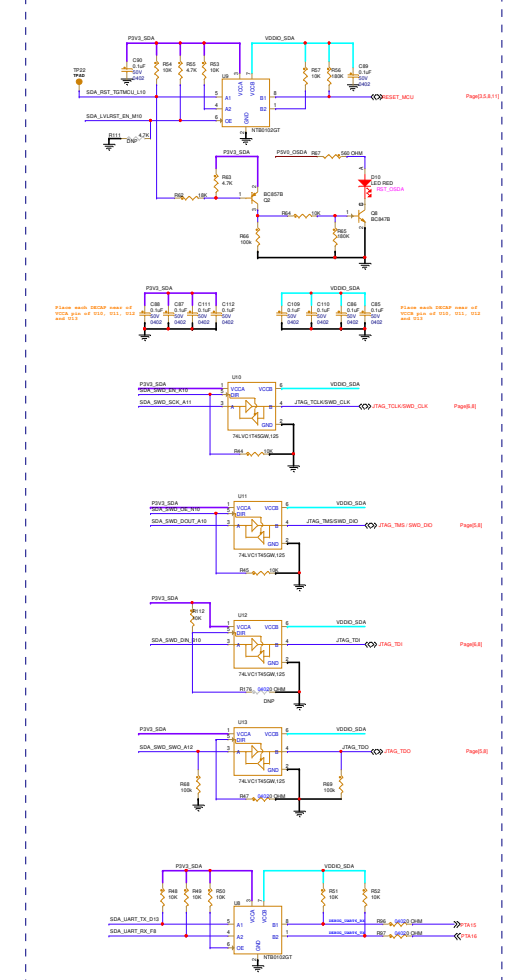
500 mA limited



OSDA JTAG Connector



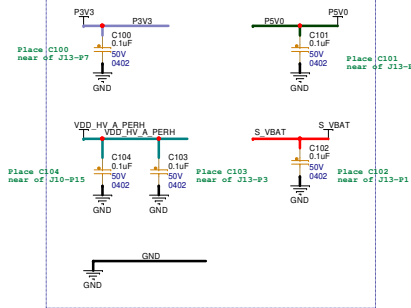
OpenSDA-MCU signals



		Automotive Product Group 6000 NXP OpenSDA Line West 6000 NXP OpenSDA Line West 6000 NXP OpenSDA Line West	
Part Number: SS2K312MINI-EVB		Revision: 1.0	
Date: 2023-01-10		OpenSDA Interface	
Author: SCHWABER, PROF. DR. RALF		Date: 2023-01-10	

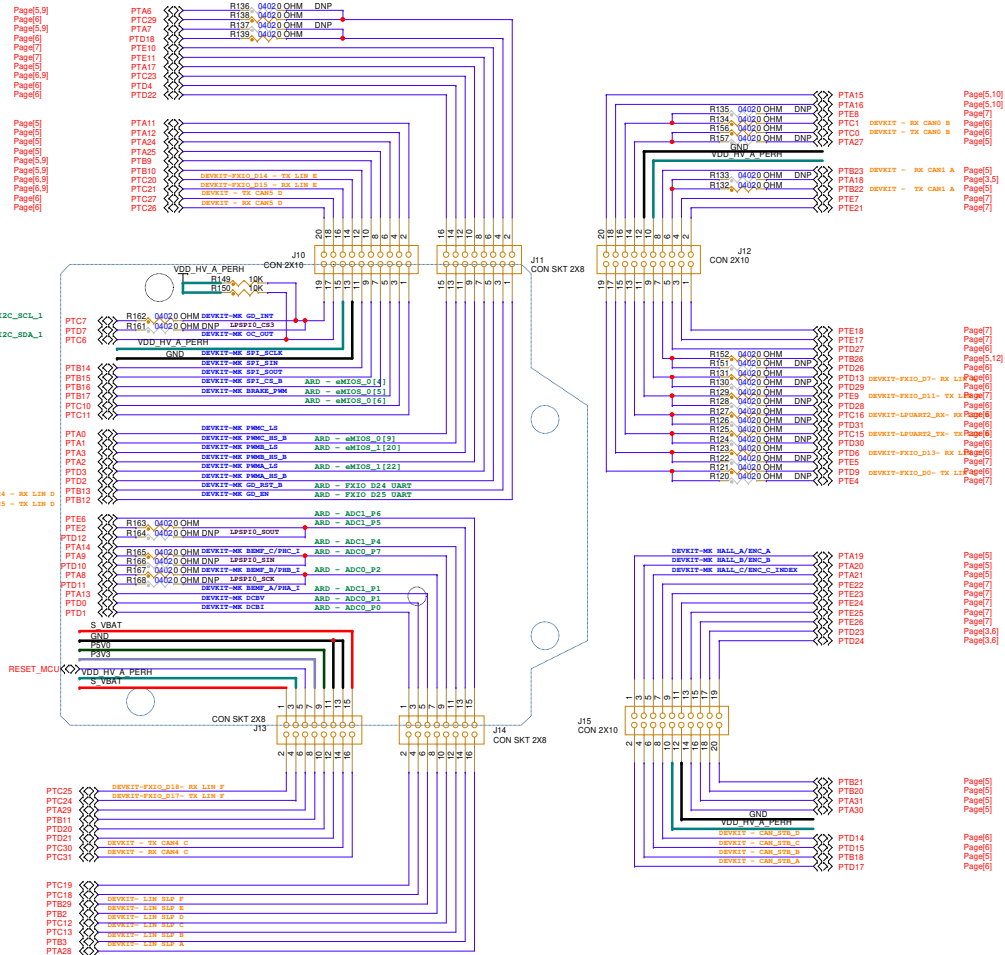
Arduino Headers

Voltage References



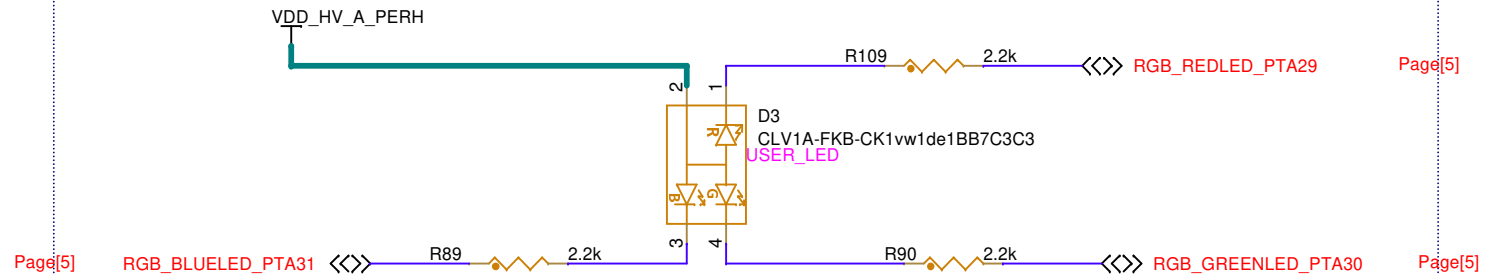
Boards compatibility:

- DEVKIT-MOTORGD --> SPF-29379
- DEVKIT-COMM --> SPF-29209
- ARDUINO UNO

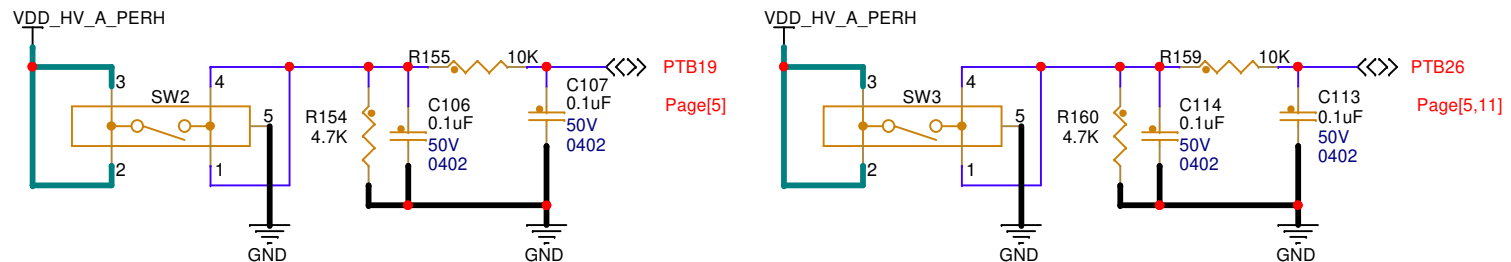


User Peripherals

RGB LED



USER BUTTON



Automotive Product Group

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